

Quiz 3: Conditional Expectation, Entropy, Stochastic Processes

Homework for lectures 5 and 6. Notation as in the lectures. Problems are numbered A1–A21, B1–B22, C1–C22; within each group they get harder, and the last one or two of a group need the concept of that group.

Part A: Conditional Expectation

Notation as in the Conditional Expectation lecture. Two dice are Y (first), Z (second), $X = Y + Z$: the two-dice table, rows the events $\{Y = y\}$. Here also $M = \max(Y, Z)$ and $D = Y - Z$.

Conditional expectation given a value

Lecture battery 1

- A1. Two dice. Which cells form the event $\{M = m\}$, and how many are there? Compute $\mathbb{E}[Z \mid M = m]$ for $m = 1, \dots, 6$ (a formula in m), then $\mathbb{E}[X \mid M = m]$ by symmetry, and deduce $\mathbb{E}[\min(Y, Z) \mid M = m]$ without further counting.
- A2. Let Y be uniform on $[1, 2]$ and, given $Y = y$, let X be uniform on $[0, y]$. Sketch the region where $f_{X,Y} > 0$ and write down $f_{X,Y}$ and $f_{X|Y}$ with their domains. Compute $\mathbb{E}[X \mid Y = y]$ and $\mathbb{E}[X^2 \mid Y = y]$.

Law of total expectation

Lecture battery 2

- A3. Roll a die, getting N ; then draw N independent numbers uniform on $[0, 1]$ and let L be the largest of them. Compute $\mathbb{E}[L]$ by conditioning on N . (Given $N = n$, $\mathbb{P}(L \leq t \mid N = n) = t^n$, so L has conditional pdf nt^{n-1} on $[0, 1]$.) Then find F_L head-on and check $\mathbb{E}[L] = \int_0^1 (1 - F_L(t)) dt$.
- A4. With Y, X as in A2, compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ by total expectation and deduce $\text{Var}(X)$. Then find f_X head-on (it is given by two different formulas on $[0, 1]$ and on $[1, 2]$) and check $\mathbb{E}[X]$.
- A5. A supplier's defect rate is unknown: model it as Θ uniform on $[0, \frac{1}{5}]$. Given $\Theta = \theta$, the number D of defective items in a sample of 10 is binomial: $\mathbb{E}[D \mid \Theta = \theta] = 10\theta$ and $\mathbb{E}[D^2 \mid \Theta = \theta] = 10\theta(1 - \theta) + 100\theta^2$. Compute $\mathbb{E}[D]$ and $\text{Var}(D)$, and compare $\text{Var}(D)$ with the variance $10 \cdot 0.1 \cdot 0.9$ of a binomial with the average defect rate 0.1. Then compute $\mathbb{P}(D = 0)$ head-on and compare with 0.9^{10} .

Random sums

Lecture battery 5

- A6. A bus carries N passengers, $\mathbb{E}[N] = 25$, the count independent of who the passengers are. Each passenger is an adult (fare £1.50) with probability 0.6 or a child (fare £0.75) with probability 0.4. (a) Find the expected takings. Does the answer change if the passengers' types are dependent on one another (families travel together)? (b) Now suppose that with probability $\frac{1}{4}$ the bus is a school run, in which case $N = 40$ and every passenger is a child; otherwise $N = 20$ and every passenger is an adult. Compute the expected takings by

conditioning on the kind of run. Then compute $\mu = \mathbb{E}[X_i]$ for a fixed passenger $i \leq 20$ and $\mathbb{E}[N]$, and explain why $\mu \mathbb{E}[N]$ is not the answer.

- A7. Two dice X_1, X_2 are rolled in turn, $\mu = \mathbb{E}[X_i] = 3.5$. In each of the following, S is the sum of the terms that are counted; compute $\mathbb{E}[S]$ by conditioning on the value of the die that decides N , and compare with $\mu \mathbb{E}[N]$. (a) Looking back: $N = 2$ if $X_1 \leq 3$, otherwise $N = 1$ (the second die is rolled and counted only when the first is small). (b) Looking ahead: $N = 2$ if $X_2 \leq 3$, otherwise $N = 1$ (both dice are rolled; the second is counted only when it is small).

Waiting times

Lecture battery 5

- A8. Let T be the number of rolls of a fair die until two sixes in a row first appear. Show $\mathbb{E}[T] = 42$ by conditioning on the opening (first roll not a six; six then not a six; six, six), as the lecture did for HH.
- A9. Toss a fair coin until the pattern THH first appears; then the same for THT and for TTT. Compute the three expected waiting times by conditioning on the first toss from each of the three states “no progress”, “T”, “TH” (respectively “TT”). Which pattern arrives soonest, and why are three patterns of the same length not equally quick?
- A10. Roll a fair die repeatedly. (a) Let T be the number of rolls until both a 1 and a 6 have appeared. Compute $\mathbb{E}[T]$ by conditioning on the first roll. (b) Let T_6 be the number of rolls until every face has appeared at least once. Writing m_k for the expected number of further rolls when k distinct faces have been seen, derive $m_k = \frac{6}{6-k} + m_{k+1}$ and compute $\mathbb{E}[T_6]$. Why is $\mathbb{P}(T_6 = n)$ unpleasant to write down?

Conditional expectation as a random variable

Lecture battery 3

- A11. Two dice, $M = \max(Y, Z)$, and $\mathbb{E}[X \mid M]$ from A1. Write down $\mathbb{E}[X \mid M](\omega)$ and $X(\omega)$ for $\omega = (2, 5), (5, 5), (5, 2), (6, 1)$. On which set of outcomes does $\mathbb{E}[X \mid M]$ take the value $\frac{70}{9}$? Describe the shape of the “rows” that conditioning on M paints, and check that they partition the table.
- A12. Two dice. Compute $\mathbb{E}[M \mid X = x]$ and $\mathbb{E}[\min(Y, Z) \mid X = x]$ for $x = 4, 7, 10$, by listing the cells of each diagonal. Verify $\mathbb{E}[M \mid X] + \mathbb{E}[\min(Y, Z) \mid X] = X$ and explain from the properties of conditional expectation why this holds without any listing. Then write $\mathbb{E}[|D| \mid X]$ in terms of $\mathbb{E}[M \mid X]$ and X .
- A13. Let U be uniform on $\{1, 2, 3, 4\}$, $V = \mathbf{1}_{\{U \text{ odd}\}}$ and $W = \mathbf{1}_{\{U \geq 3\}}$. List the values of the four random variables $\mathbb{E}[U \mid V]$, $\mathbb{E}[U \mid W]$, $\mathbb{E}[V \mid W]$, $\mathbb{E}[W \mid V]$ at each of the four outcomes, saying which rows each conditioning creates. Two of the four are constant: what does that suggest about V and W , and is it true?

The tower property

Lecture battery 4

- A14. Let N be Poisson with mean λ , that is, $\mathbb{P}(N = n) = e^{-\lambda} \lambda^n / n!$ for $n = 0, 1, 2, \dots$, so that $\mathbb{E}[N] = \lambda$. Toss a fair coin N times; H is the number of heads. Compute $\mathbb{E}[H]$. Then show that H is Poisson with mean $\lambda/2$, and compare the effort.
- A15. A stick of length 1 is broken at a uniform point Y ; the longer piece is kept. Let L be its length. Compute $\mathbb{E}[L]$ two ways: directly from the definition of L , and by conditioning on the event $\{Y < \frac{1}{2}\}$. Then: the kept piece is broken again at a uniform point, and the longer part kept, with length L_2 . Compute $\mathbb{E}[L_2]$.

Taking out what is known

Lecture battery 4

- A16. Two dice, $M = \max(Y, Z)$. Compute $\mathbb{E}[MZ]$ using $\mathbb{E}[h(M)Z] = \mathbb{E}[h(M)\mathbb{E}[Z | M]]$ with $\mathbb{E}[Z | M]$ from A1 and the pmf $p_M(m) = \frac{2m-1}{36}$. Deduce $\text{Cov}(M, Z)$. Check $\mathbb{E}[MZ]$ against the table.
- A17. With Y, X as in A2 (Y uniform on $[1, 2]$, X uniform on $[0, Y]$ given Y), compute $\mathbb{E}[XY]$ and $\mathbb{E}[XY^2]$ by taking out what is known, and deduce $\text{Cov}(X, Y)$. Which double integrals did you avoid?

Independence

Lecture battery 4

- A18. Two dice, $D = Y - Z$. Compute $\mathbb{E}[D | Y]$, $\mathbb{E}[D | Z]$ and $\mathbb{E}[D | X]$ as random variables. Which of the three conditionings carries information about D , and which property of conditional expectation settles each one? (For $\mathbb{E}[D | X]$, look at how the diagonal $\{X = x\}$ is placed relative to the main diagonal.)
- A19. Two dice, $D = Y - Z$, $X = Y + Z$. Show $\mathbb{E}[X | D] = 7 = \mathbb{E}[X]$ (which cells form $\{D = d\}$, and what is X on them?), and recall $\mathbb{E}[D | X] = 0 = \mathbb{E}[D]$ from A18. Deduce $\text{Cov}(X, D) = 0$. Then show that X and D are *not* independent, by comparing the parities of X and D or by computing $\mathbb{P}(D = 5, X = 7)$ against $\mathbb{P}(D = 5)\mathbb{P}(X = 7)$. Which lesson of the lecture is this an instance of?

Putting it together

Lecture battery 5

- A20. A gambler bets on a fair coin and doubles her stake after each loss (stakes $\pounds 1, 2, 4, \dots$), stopping after the first win; R is the number of rounds. (a) Show that her net gain is exactly $\pounds 1$ on every outcome, and compute the expected total amount staked by conditioning on R . (b) Now the table limit stops her after round 5 whether or not she has won. Compute her net gain on each outcome, then $\mathbb{E}[\text{net gain}]$, $\mathbb{E}[R]$ and the expected total staked. What do you conclude about the doubling strategy?
- A21. Let $X_1, X_2, \dots$ be independent, each uniform on $\{1, \dots, 6\}$, and let N be the index of the first roll showing 5 or 6; N is determined by the X_i and not independent of them. (a) Let $S = X_1 + \dots + X_N$ include the stopping roll. Compute $\mathbb{E}[S]$ by conditioning on the first roll, and compare with $\mu \mathbb{E}[N]$. (b) Let $S' = X_1 + \dots + X_{N-1}$ exclude it. Compute $\mathbb{E}[S']$ the same way and compare with $\mu \mathbb{E}[N - 1]$. Explain the two comparisons.

Part B: Entropy

Logarithms are base 2, the unit is the bit, and $0 \log 0 = 0$. Notation as in the Entropy lecture: p_X is the pmf of X and S_X its support; $I_X(x) = \log \frac{1}{p_X(x)}$; $H(X)$ is entropy and $h(p)$ the binary entropy function; $H(P, Q)$ is the cross-entropy of the truth P with the code or model Q , $D(P \| Q) = H(P, Q) - H(P)$; $H(X | Y)$ and $I(X; Y)$ are conditional entropy and mutual information. Two dice are always Y (first), Z (second), $X = Y + Z$, the two-dice table.

Codes and Kraft's inequality

Lecture Section 1

- B1. A source emits one of five symbols a, b, c, d, e with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}$. (a) Check that the lengths $(1, 2, 3, 4, 4)$ satisfy Kraft's inequality with equality and write down a prefix-free code c with these lengths. (b) Compute $L(c)$ and $H(X)$. (c) Decode 111001011111100 with your code. (d) Compute $\text{Var}(\ell_c(X))$ and the mean and standard deviation of the number of bits needed for 1000 symbols.

- B2. (a) Which of the following can be the lengths of a prefix-free code: $(1, 2, 3, 3, 4)$ for five symbols; $(2, 2, 2, 3, 3, 3)$ for six; $(2, 2, 3, 3, 3, 3)$ for six; $(1, 3, 3, 3, 3)$ for five; $(1, 2, 3, 4, 5, 5)$ for six? For each admissible profile write down a code. (b) Show that if a prefix-free code has $\sum_i 2^{-\ell_i} < 1$, then there is a bit string that is neither a prefix of a codeword nor has a codeword as a prefix, so a further symbol could be added to the code without lengthening any existing codeword.
- B3. Optimality of the rover's code. Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$. (a) Using Kraft's inequality and Gibbs' lemma, show that every prefix-free code for four symbols has $L(c) \geq 1.75$ under p , so $0, 10, 110, 111$ is optimal. (b) Show that the only length profiles for four symbols with Kraft sum exactly 1 are $(2, 2, 2, 2)$ and $(1, 2, 3, 3)$, up to order. (Consider the shortest length.) (c) Take as given that an optimal code has Kraft sum 1 (otherwise a codeword can be shortened). Deduce that for *every* pmf on four symbols the optimal code is either the fixed-length code or $0, 10, 110, 111$ with the symbols assigned in decreasing order of probability, whichever has the smaller L . (d) Apply (c) to $p = (0.45, 0.25, 0.15, 0.15)$ and to $p = (0.3, 0.3, 0.2, 0.2)$, and compare L^* with H in each case. Why can the source coding theorem alone not decide which code is optimal here?

Information

Lecture Section 2

- B4. Let $f: (0, 1] \rightarrow [0, \infty)$ be non-increasing with $f(pq) = f(p) + f(q)$ and $f(\frac{1}{2}) = 1$. (a) Compute $f(\frac{1}{4})$, $f(\frac{1}{64})$ and $f(2^{-1/3})$. (b) Using $2^7 < 3^5 < 2^8$, show $\frac{7}{5} \leq f(\frac{1}{3}) \leq \frac{8}{5}$. (c) Using $2^{19} < 3^{12} < 2^{20}$, sharpen this to $\frac{19}{12} \leq f(\frac{1}{3}) \leq \frac{20}{12}$, and compare with $\log 3 = 1.585$. (d) $g(p) = \frac{1}{p} - 1$ is decreasing with $g(\frac{1}{2}) = 1$. Which requirement does it fail, and at which numbers?
- B5. Two dice. Let $A = \{\text{at least one six}\}$, $B = \{Y = Z\}$, $C = \{Y \text{ even}\}$, $D = \{X \text{ odd}\}$, $E = \{X \geq 10\}$, $F = \{Y = 6\}$. (a) Compute $I(A)$, $I(B)$ and $I(A \cap B)$. Do the informations add? Why not? (b) Compute $I(C)$, $I(D)$ and $I(C \cap D)$, and explain the result by listing the cells of $C \cap D$. (c) Compute $I(E)$, $I(F)$ and $I(E \cap F)$. Check that $I(E \cap F) = I(F) + \log \frac{1}{\mathbb{P}(E|F)}$ and say in words why $E \cap F$ is less surprising than $I(E) + I(F)$ would suggest.
- B6. The rover's source, $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$, coded with $0, 10, 110, 111$. Let $L = \ell_c(X)$ be the codeword length of one instruction. (a) Write down the pmf of L and compute $\mathbb{E}[L]$ and $\text{Var}(L)$. (b) For 1000 instructions, give the mean and standard deviation of the total number of bits, and compare with the fixed-length code. (c) Show that a stream of independent fair bits cannot be compressed: any prefix-free code for blocks of k bits has average length at least k . Why does it follow that compressing the rover's coded stream a second time gains nothing?

Entropy

Lecture Section 3

- B7. Compute the following entropies. (a) $H(0.4, 0.3, 0.2, 0.1)$. (b) N = the number of heads in three tosses of a fair coin. Compute $H(N)$ and compare with the entropy of the three tosses themselves. Where did the missing bits go? (Compute $H(\text{tosses} | N)$.) (c) $M = \min(Y, Z)$ for two dice. Compute $H(M)$ from its pmf and compare it with $H(Y)$ and with $H(\max(Y, Z))$.
- B8. Coin A shows heads with probability 0.9, coin B is fair. One coin is chosen uniformly at random ($C \in \{A, B\}$) and tossed once, giving $X \in \{H, T\}$. (a) Compute the pmf of X and $H(X)$. (b) Compute $H(X | C = A)$, $H(X | C = B)$ and $H(X | C)$. (c) Compare $H(X)$ with $H(X | C)$ and state the property of h that this illustrates. What is $I(X; C)$?

(d) Compute $\mathbb{P}(C = A \mid X = H)$ and $\mathbb{P}(C = A \mid X = T)$, then $H(C \mid X)$, and check $H(C) - H(C \mid X) = I(X; C)$.

- B9. Roll a fair die until the first six; N is the number of rolls, so $\mathbb{P}(N = k) = (\frac{5}{6})^{k-1} \frac{1}{6}$ for $k \geq 1$. (a) Write $I_N(k)$ as a function of k . (b) Compute $H(N) = \mathbb{E}[I_N(N)]$ without summing a series, using $\mathbb{E}[N] = 6$. (c) Show that for a geometric variable with success probability p , $H(N) = h(p)/p$, and check the case $p = \frac{1}{2}$. (d) Compute $H(\min(N, 6))$, the entropy of reporting the value only up to “6 or more”, and compare.

The source coding theorem

Lecture Section 4

- B10. Let $p = (0.5, 0.2, 0.2, 0.1)$. (a) Compute $H(p)$. (b) Compute Shannon’s lengths $\lceil \log \frac{1}{p(x)} \rceil$, their Kraft sum, a code with these lengths, and its average length. (c) Using B3(c), find the optimal code and L^* . (d) Verify $H \leq L^* \leq L_{\text{Shannon}} < H + 1$. Which of these would be an equality if fractional lengths were allowed?
- B11. A fair die is rolled repeatedly and the results are to be transmitted. Take as given that for M equiprobable symbols the optimal prefix-free code uses only the lengths $\lfloor \log M \rfloor$ and $\lceil \log M \rceil$ and has Kraft sum 1. (a) Single rolls, $M = 6$: find how many codewords have each length and compute L^* per roll. (b) Pairs of rolls, $M = 36$: the same; give the cost per roll. (c) Triples, $M = 216$: the same. (d) Compare with Shannon’s construction (all lengths $\lceil \log M \rceil$) for each block size, and with the floor $\log 6 = 2.585$. Which theorem guarantees that the cost per roll tends to 2.585?

Cross-entropy and relative entropy

Lecture Section 5

- B12. $P = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ and $Q = (\frac{1}{8}, \frac{1}{8}, \frac{3}{4})$ on $\{a, b, c\}$. (a) Compute $H(P)$ and $H(Q)$. (b) Compute $H(P, Q)$ and $D(P \parallel Q)$. (c) Compute $H(Q, P)$ and $D(Q \parallel P)$. (d) Say in words which code is used on which source in (b) and in (c). (e) P ’s ideal lengths $(1, 2, 2)$ are integers, Q ’s are not. Find the best prefix-free code for Q (B3(b) for three symbols: the only complete profile is $(1, 2, 2)$), its cost on Q , and its cost on P . Why is the last number smaller than $H(P, Q)$?
- B13. (a) $P = (0.9, 0.05, 0.05)$ and $U = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Compute $H(P)$, $H(P, U)$, $D(P \parallel U)$, $H(U, P)$ and $D(U \parallel P)$. The lecture exercises gave $D(P \parallel U) = \log n - H(P)$ for the uniform U on n symbols; show that the other direction is $D(U \parallel P) = -\log n - \frac{1}{n} \sum_x \log P(x)$, and say which of the two is unbounded as P moves towards the boundary of Δ_n . (b) Two coins, $A = (0.99, 0.01)$ and $B = (0.9, 0.1)$. Compute $D(A \parallel B)$ and $D(B \parallel A)$ and explain the asymmetry in one sentence each way. (c) A model Q is to be scored against the truth P . Which of $D(P \parallel Q)$, $D(Q \parallel P)$ measures the bits it wastes, and why must the truth be the first argument?
- B14. The rover has two missions, equally likely: $P_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ and $P_2 = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ for (up, down, left, right). One code, built for a pmf Q , must be installed before the mission is known; its cost is $\frac{1}{2}H(P_1, Q) + \frac{1}{2}H(P_2, Q)$ bits per instruction. (a) Show that this cost equals $H(\bar{P}, Q)$ with $\bar{P} = \frac{1}{2}(P_1 + P_2)$, and compute $\bar{P}$. (b) Find the Q that minimises the cost and the minimal cost. Compare with keeping the old code ($Q = P_1$), with the fixed-length code, and with the 1.75 bits achievable if the mission were known. (c) Show that the average waste $\frac{1}{2}D(P_1 \parallel \bar{P}) + \frac{1}{2}D(P_2 \parallel \bar{P})$ equals $I(M; X)$, the mutual information between the mission $M \in \{1, 2\}$ and one instruction X , and compute it. (d) Codeword lengths must be integers. Using B3(b), find the best single-symbol code for the two-mission problem. What must be done to approach the cost in (b)?

The log loss

Lecture Section 6

- B15. A three-class classifier. For a certain input the true label has pmf $p = (0.6, 0.3, 0.1)$. Five models report $q_1 = (0.6, 0.3, 0.1)$, $q_2 = (0.8, 0.15, 0.05)$, $q_3 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $q_4 = (0.98, 0.01, 0.01)$, $q_5 = (0.5, 0.4, 0.1)$. (a) Compute the expected log loss $H(p, q_i)$ of each. (b) Decompose each as $H(p) + D(p \| q_i)$. (c) q_4 puts the most probability on the most likely class, yet scores worst. Explain with one number. (d) A validation loss curve for such inputs has flattened at 1.30 bits per example. Is further training worthwhile? What is the perplexity at the floor?

Wordle

Lecture Section 7

- B16. Toy Wordle with three-letter words. $W = \{\text{ART, RAT, TAR, CAT, ACT, TIN, NIT, RIB}\}$, $N = 8$, answer A uniform on W . Colours: **G** right letter in the right place, **Y** right letter in the wrong place, **B** absent. (a) Write the pattern table $w \mapsto \pi_{\text{TIN}}(w)$ for the guess **TIN**, list the buckets and compute $H(\Pi)$. (b) The same for **RIB**. (c) Show that **CAT** gives eight distinct patterns, so $H(\Pi) = 3 = \log N$, the maximum possible. (d) **TAN** is not on the list. Compute its bucket sizes and entropy. Can any guess, on the list or off it, beat **CAT**? (e) Compute the expected number of guesses $\mathbb{E}[G]$ for the strategy “**CAT**, then the word it identifies” and for “**TIN**, then **CAT** if the bucket has four words, otherwise the identified word”. Do the entropy ranking and the score ranking agree?
- B17. Real Wordle, $N = 2315$ answers, 243 patterns. (a) A guess produces only five patterns, with bucket sizes 1000, 800, 400, 100, 15. Compute $H(\Pi)$ and the expected size of the remaining list, $\mathbb{E}[|\text{bucket of } \Pi|] = \sum_{\pi} |\text{bucket}| \cdot p_{\Pi}(\pi)$. (b) Another guess splits the list as evenly as 243 patterns allow: 128 buckets of size 10 and 115 of size 9. Compute $H(\Pi)$ and the expected remaining size. (c) Is the expected remaining size equal to $2^{\log N - H(\Pi)}$? Check with (a) and (b) and explain the discrepancy. (d) After a pattern whose bucket has 39 words, how many bits of uncertainty remain, and what is the largest amount of information the next guess could deliver?

Joint and conditional entropy, mutual information

Lecture Section 8

- B18. Let X be the sum of two dice, $H(X) = 3.274$ bits. (a) The lecture exercises proved $H(g(X)) \leq H(X)$ for every function g . Show, from the chain rule for the pair $(X, g(X))$, that the gap is $H(X) - H(g(X)) = H(X | g(X))$, and say in words what this conditional entropy measures. (b) $g(X)$ = the parity of X . Compute $H(g(X))$ and $H(X | g(X))$. (c) $g(X) = X \bmod 3$. Compute $H(g(X))$ and $H(X | g(X))$ from (a). Then compute $H(X | g(X))$ directly, as the average of the three row entropies $H(X | g(X) = r)$, $r = 0, 1, 2$, and check that the two agree. (d) $g(X) = \mathbf{1}\{X \geq 7\}$. Compute $H(g(X))$. (e) Among all functions g of X with two values, which have the largest entropy, and what is it?
- B19. Two dice, $D = |Y - Z|$. (a) Write the pmf of D and compute $H(D)$. (b) Compute $H(D | Y = y)$ for each row y and hence $H(D | Y)$. (c) Compute $I(D; Y)$ and $H(Y | D)$. (d) Let $X = Y + Z$. Show that the pair (X, D) determines the unordered pair $\{Y, Z\}$ and nothing more, compute $H(X, D)$ from that, and hence $I(X; D)$. Explain where at least one of those bits comes from. (e) Compare $I(X; D)$ with $I(X; Y) = 0.689$ from the lecture.
- B20. A bit X with $\mathbb{P}(X = 1) = 0.3$ is sent down a wire that flips it with probability $\varepsilon = 0.2$, independently of X ; Y is received. (a) Write the 2×2 table of $p_{X,Y}$, rows $\{X = 0\}, \{X = 1\}$. (b) Compute $H(Y | X = 0)$, $H(Y | X = 1)$ and $H(Y | X)$. (c) Compute $\mathbb{P}(Y = 1)$, $H(Y)$ and $I(X; Y)$. (d) Compute $I(X; Y)$ for a uniform input and show that, for fixed $\varepsilon < \frac{1}{2}$, $I(X; Y)$ is maximised by the uniform input. (e) Compute $\mathbb{P}(X = 1 | Y = 1)$, $\mathbb{P}(X = 1 | Y = 0)$ and $H(X | Y)$, and check $I(X; Y) = H(X) - H(X | Y)$.

- B21. A binary text source is modelled as a Markov chain on $S = \{0, 1\}$ with transition matrix $Q = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$ (the chain of C12). (a) Find the stationary pmf π . (b) Compute the entropy rate $H(X_{i+1} | X_i)$ and the monographic entropy $H(\pi)$, and identify their difference. (c) Compute $H(X_1, X_2, X_3)$ for the chain started from π , once by the chain rule and once by listing the eight probabilities. (d) Give the perplexity of the Markov model, of the monographic model and of the uniform model. (e) What does a compressor built on the monographic model pay per symbol on this text, and how much of it is waste?
- B22. Perplexity, compression and a cipher. (a) A letter-level model scores 3.5 bits per letter on 27-symbol English. Give its perplexity and compare with Shannon's digraphic model (3.32 bits, perplexity 10). (b) A word-level model reports perplexity 40 per word. English words average 5.5 letters including the following space. Convert to bits per letter and perplexity per letter. Which of the two models is better? (c) How many kilobytes does each model need to compress a text of 10^6 letters, and how many would the uniform model ($\log 27 = 4.75$ bits) need? (d) A Vigenère cipher with key length 10 has 26^{10} keys. Compute $H(K)$ and the unicity distance for $H(\text{English}) = 1.5$ and for 1 bit per letter (redundancy $R = \log 26 - H$). Why does a longer key need a longer ciphertext to be determined?

Part C: Stochastic Processes

Notation as in the Stochastic Processes lecture: $\xi_1, \xi_2, \dots$ are the ± 1 steps, $S_n = \xi_1 + \dots + \xi_n$, and $\mathbb{P}_i, \mathbb{E}_i$ refer to a start at i . $\mathbf{P} = (p(i, j))$ is the transition matrix, row i the conditional pmf of the next state given the current state i ; $p^{(n)}(i, j)$ are the entries of $\mathbf{P}^n$; π is a stationary distribution; T_A is the hitting time of A , $h_i^A = \mathbb{P}_i(T_A < \infty)$, $k_i^A = \mathbb{E}_i[T_A]$; $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$ is the fundamental matrix. Φ is the $N(0, 1)$ cdf; values used below:

x	0.100	0.500	0.577	0.674	0.707	0.905	1.000	1.155	1.225	
$\Phi(x)$	0.540	0.691	0.718	0.750	0.760	0.817	0.841	0.876	0.890	$\Phi(1.414) = 0.921$.

Processes and the random walk

Lecture batteries 1–2

- C1. Three fair coin tosses, $\xi_t = \pm 1$, $S_t = \xi_1 + \dots + \xi_t$: the eight-row table of the lecture. (a) Write the joint pmf of (S_1, S_3) as a table and check that S_1 and S_3 are not independent. (b) Read $\mathbb{E}[S_3 | S_1 = s]$ and $\mathbb{E}[S_3 | S_2 = s]$ off the rows, for every s , and explain the answers by independent increments. (c) Compute $\mathbb{E}[S_3^2 | S_2 = s]$ for $s = -2, 0, 2$ and write $\mathbb{E}[S_3^2 | S_2]$ as a random variable.
- C2. A walk with $p = 0.55$ (a slightly favourable game), $n = 100$ steps. (a) Compute $\mathbb{E}[S_{100}]$, $\text{Var}(S_{100})$ and the standard deviation. (b) Compute $\mathbb{P}(S_{100} = 10)$ and $\mathbb{P}(S_{100} = 11)$. (c) Approximate $\mathbb{P}(S_{100} > 0)$ by the normal distribution (use $\{S_{100} \geq 2\}$ and the midpoint 1 as the cut). (d) Compute $\text{Cov}(S_{30}, S_{100})$ and the correlation of S_{30} and S_{100} . (e) Fair walk: compute $\mathbb{P}(S_8 = 0)$, $\mathbb{P}(S_8 = 4)$ and $\mathbb{P}(S_9 = 0)$.
- C3. Fair walk, $M_8 = \max_{k \leq 8} S_k$. (a) Compute $\mathbb{P}(M_8 \geq 2)$ by the reflection principle. (b) Compute $\mathbb{P}(M_8 = 0)$. (c) Tabulate $\mathbb{P}(M_8 \geq m)$ for $m = 1, \dots, 8$ and deduce the pmf of M_8 . (d) Compute $\mathbb{E}[M_8] = \sum_{m \geq 1} \mathbb{P}(M_8 \geq m)$. What would the head-on computation require?
- C4. A fair walk on $\{0, 1, \dots, a\}$ with a reflecting wall: from 0 the next position is 1 with probability one; from $0 < i < a$ it moves to $i \pm 1$ with probability $\frac{1}{2}$ each; a is absorbing. Let $m_i = \mathbb{E}_i[\text{time to reach } a]$. (a) Write the first-step equations for $m_0, \dots, m_a$. (b) Solve

them for $a = 3$. (c) Show that $m_i = a^2 - i^2$ for every a , and explain the agreement with the lecture's exit time from $(-a, a)$.

Brownian motion

Lecture battery 3

- C5. Let B be a standard Brownian motion. (a) Give the distribution of $B_3 - B_1$ and compute $\mathbb{P}(B_3 - B_1 > 2)$. (b) Compute $\mathbb{P}(B_4 > 1 \mid B_1 = 2)$ and $\mathbb{P}(B_4 > 1 \mid B_1 = -1)$ by splitting into increments. (c) Write $\mathbb{E}[B_4 \mid B_1]$ and $\mathbb{E}[B_4^2 \mid B_1]$ as random variables. (d) Now condition the other way. Explain by symmetry between the three increments $B_1, B_2 - B_1, B_3 - B_2$ why $\mathbb{E}[B_1 \mid B_3] = B_3/3$. Then, by completing the square in the joint pdf of (B_1, B_3) , show that given $B_3 = x$, $B_1 \sim N(\frac{x}{3}, \frac{2}{3})$, and compute $\mathbb{P}(B_1 > 0 \mid B_3 = 3)$.
- C6. Use $\text{Cov}(B_s, B_t) = \mathbb{E}[B_s B_t] = \min(s, t)$ throughout. (a) Compute $\text{Var}(B_1 + B_2 + B_3)$. (b) Compute $\text{Var}(B_2 - 2B_1)$ and $\text{Cov}(B_2 - 2B_1, B_1)$. For which constant c is $B_2 - cB_1$ uncorrelated with B_1 ? Which property of Brownian motion says that for this c it is in fact independent of B_1 ? (c) Show $\text{Corr}(B_s, B_t) = \sqrt{s/t}$ for $s < t$, and evaluate it for $(1, 4)$ and $(1, 100)$. (d) Let $X_t = B_{t+1} - B_t$ for $t \geq 0$. Show that $X_t \sim N(0, 1)$ for every t and that $\text{Cov}(X_s, X_t) = \max(0, 1 - |t - s|)$. In what sense is X stationary although B is not?
- C7. The reflection principle for Brownian motion: $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a)$. Let T_1 be the first time B reaches 1. (a) Compute $\mathbb{P}(\max_{s \leq 4} B_s \geq 2)$. (b) Write $\mathbb{P}(T_1 \leq t)$ in terms of Φ and evaluate it at $t = 1, 4, 100$. (c) Find the median of T_1 (the t with $\mathbb{P}(T_1 \leq t) = \frac{1}{2}$). (d) By scaling, what is the median of T_a ? (e) The lecture showed $\mathbb{E}[T_1] = \infty$. Reconcile this with a median of about 2.

Risk of ruin

Lecture batteries 4–5

- C8. A gambler with £20 plays £1 rounds until she has £50 or nothing. (a) Fair game: compute the probability of reaching £50 and the expected number of rounds. (b) $p = 0.48$: the same, to three decimals. (c) She switches to £10 rounds. Redo (a) and (b) (a walk from 2 to 5); verify the unfair probability by solving the four first-step equations directly rather than by the formula. (d) State in one sentence what the stake changes and what it does not.
- C9. Red at roulette: European wheels have one zero ($p = \frac{18}{37}$), American wheels two ($p = \frac{18}{38}$). A player starts with £100 and bets £1 a spin. (a) Compute the probability of reaching £200 on each wheel, and the ratio. (b) Compute the expected number of spins on each wheel. (c) European wheel, modest target £110: compute h_{100} and $\mathbb{E}[T]$, and check the identity mean gain $= Nh_i - i = (p - q)\mathbb{E}_i[T]$ numerically. (d) With no target, ruin is certain on both wheels. Compute the expected number of spins until ruin on each.
- C10. Bold play. $p = 0.4$, fortune £3, target £10. (a) Timid play (£1 per round): compute h_3 and $\mathbb{E}_3[T]$. (b) Bold play: each round stake $\min(i, 10 - i)$, so the fortune moves from 3 to 6 or 0, from 6 to 10 or 2, from 2 to 4 or 0, from 4 to 8 or 0, from 8 to 10 or 6. Draw the chain on $\{0, 2, 3, 4, 6, 8, 10\}$ and write the hitting system for $h_i = \mathbb{P}_i(\text{reach } 10)$. (c) Solve it and compare h_3 with (a). (d) Compute the expected number of bets under bold play from £3. (e) Repeat (c) for the fair game $p = \frac{1}{2}$. Why is the answer the same as for timid play there?

- C11. A taxi works three zones A, B, C. The zone where a trip ends, hence where the next trip starts, depends only on where the current trip started:

$$\mathbf{P} = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{pmatrix} \quad (\text{rows and columns in the order A, B, C}).$$

X_n is the starting zone of trip n . (a) Check that $\mathbf{P}$ is stochastic and say in words what row B means. (b) Compute $\mathbb{P}(X_1 = \text{C}, X_2 = \text{B} \mid X_0 = \text{A})$. (c) Compute row A of $\mathbf{P}^2$ by hand and read off $\mathbb{P}(X_2 = \text{C} \mid X_0 = \text{A})$. Which three two-trip sequences does that entry add up? (d) X_0 is A or B with probability $\frac{1}{2}$ each. Compute the distributions of X_1 and X_2 . (e) Given $X_1 = \text{B}$, does the value of X_0 change the distribution of X_2 ? Which property says so?

- C12. The two-state chain $\mathbf{P} = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$ on $\{0, 1\}$ (the text source of B21). (a) Find the eigenvalues of $\mathbf{P}$ and the stationary distribution π . (b) Show, by induction or from the lecture's formula for two-state chains, that $\mathbf{P}^n = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{4} \end{pmatrix} + 0.6^n \begin{pmatrix} \frac{1}{4} & -\frac{1}{4} \\ -\frac{3}{4} & \frac{3}{4} \end{pmatrix}$, and check $n = 1, 2$. (c) Compute $p^{(3)}(0, 0)$ and $\mathbb{P}(X_3 = 0 \mid X_0 = 1)$ exactly. (d) Find the smallest n for which every entry of $\mathbf{P}^n$ is within 0.01 of π .

- C13. The vowel/consonant chain of both lectures: $S = \{\text{V}, \text{C}\}$, $\mathbf{P} = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}$, $\pi = (\frac{5}{13}, \frac{8}{13})$. (a) Compute $\mathbf{P}^2$ and $\mathbf{P}^3$. (b) Compute the expected number of vowels among the next three letters, starting after a consonant and starting after a vowel, by linearity column by column; compare with the long-run value $3\pi_{\text{V}}$. (c) Compute $\mathbb{P}(X_1 = \text{V}, X_2 = \text{V} \mid X_0 = \text{C})$ and compare with the stationary probability of the digraph VV used in the Entropy lecture. (d) Compute the expected number of letters until the first consonant, starting from V; then the expected number of letters until the pattern VC is first completed, from V and from C. (e) Compute the mean return time to C from π and confirm it by a first-step argument.

Stationary distributions

Lecture battery 8

- C14. Random walks on graphs: $p(i, j) = 1/d(i)$ for each neighbour j of i . (a) The path $0-1-2-3$. Find π by detailed balance and verify the balance equation on the edges $(0, 1)$ and $(1, 2)$. (b) The star with centre 0 and leaves 1, 2, 3. Find π ; compute the mean return time to a leaf from π and confirm it by a first-step argument. (c) The Ehrenfest urn with $N = 3$ balls. Find π by detailed balance. (d) The biased walk on the cycle $\mathbb{Z}_3$: $p(i, i+1) = \frac{2}{3}$, $p(i, i-1) = \frac{1}{3}$. Show that the uniform distribution is stationary but that detailed balance fails. What does the chain do that a reversible chain cannot?
- C15. The taxi chain of C11. (a) Solve $\pi\mathbf{P} = \pi$, $\sum_i \pi_i = 1$ exactly. (b) $\mathbf{P}^{10}$ has every row equal to $(0.3214, 0.4286, 0.2500)$ to four places. Confirm that this agrees with (a). (c) Test detailed balance on the pair (A, B). Is the chain reversible? (d) The chain is irreducible and aperiodic. What do the convergence theorem and the ergodic theorem each say about $\pi_{\text{C}} = \frac{1}{4}$?
- C16. PageRank with damping $d = 0.85$. (a) Three pages: $\text{A} \rightarrow \text{B}$, $\text{A} \rightarrow \text{C}$, $\text{B} \rightarrow \text{C}$, $\text{C} \rightarrow \text{A}$. Write $\mathbf{H}$ and $\mathbf{P} = 0.85\mathbf{H} + 0.05\mathbf{J}$. (b) Solve $\pi\mathbf{P} = \pi$ and rank the pages. A and B each have exactly one in-link; why does A rank higher? (c) Find π without damping ($d = 1$). (d) Add

a page D with links $C \rightarrow D$ (so C now links to A and D) and $D \rightarrow D$ only. Find π without damping, and with $d = 0.85$. What does D do to the surfer, and how much does damping help?

Classes, periods, recurrence

Lecture battery 9

C17. A chain on $S = \{1, \dots, 6\}$ with

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0.2 & 0 & 0.4 & 0 & 0.4 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0.5 & 0.5 \end{pmatrix}.$$

(a) Find the communicating classes, say which are closed, give the period of each class, and classify each state as recurrent or transient. (b) Compute the probability that the chain started at 3, and at 4, ends up in the class $\{1, 2\}$. (c) Compute the expected time to leave $\{3, 4\}$ from 3 and from 4. (d) Find all stationary distributions. (e) Started at 3, does the distribution of X_n converge? Describe its behaviour for large n .

C18. Walks on the cycle $\mathbb{Z}_L = \{0, \dots, L-1\}$, addition modulo L . (a) $L = 6$, steps ± 1 with probability $\frac{1}{2}$ each. Compute $p^{(n)}(0, 0)$ for $n = 1, \dots, 4$, give the period, and describe the rows of $\mathbf{P}^n$ for large n (is π stationary; do the rows converge to it?). (b) $L = 7$, steps ± 1 . Give the period and justify it with two round-trip lengths. (c) $L = 6$, steps $+1$ or $+2$ with probability $\frac{1}{2}$ each. Is the chain irreducible? Aperiodic? What is π , and do the rows of $\mathbf{P}^n$ converge? (d) $L = 6$, steps $+2$ or $+4$ with probability $\frac{1}{2}$ each. Find the communicating classes and the period within each. (e) $L = 4$, step $+1$ with probability one. Find π , the period, and say which of the two long-run theorems applies.

Hitting probabilities and times

Lecture battery 10

C19. Tennis at deuce. From deuce the server wins each point with probability p , independently. States: D (deuce), I (advantage server), O (advantage receiver), W (game won), L (game lost). From D the next state is I or O; from I it is W or D; from O it is D or L. (a) Write the transition matrix. (b) Write the hitting system for $h_i = \mathbb{P}_i(\text{reach W})$ and solve it; show $h_D = \frac{p^2}{p^2 + q^2}$. (c) Write the system for $k_i = \mathbb{E}_i[\text{points until the game ends}]$ and show $k_D = \frac{2}{1 - 2pq}$. (d) Evaluate h_D and k_D for $p = 0.5, 0.55, 0.6, 0.7$. What does a 55/45 edge per point become per game? (e) What is the head-on computation of h_D , and why does the first-step system avoid it?

C20. The taxi chain of C11 again. (a) Compute the expected number of trips until the next trip starting in C, from A and from B. (b) Compute the mean return time to C by first-step analysis and compare with $1/\pi_C$. (c) Make C absorbing and compute the fundamental matrix $\mathbf{N}$; recover (a) from $\mathbf{N}\mathbf{1}$ and read off the expected number of trips starting in B before the next C-start, when the current trip starts in A.

Convergence, the ergodic theorem, absorbing chains

Lecture batteries 11–12

C21. Long-run behaviour of the taxi chain of C11 and C15. (a) The eigenvalues of $\mathbf{P}$ are 1, 0.3 and 0.2 (check the trace and the determinant). How fast do the rows of $\mathbf{P}^n$ approach π ?

The maximal deviations are 0.179, 0.051, 0.015 at $n = 1, 2, 3$; compare with 0.3^n . (b) State what fraction of trips start in each zone in the long run, and which theorem says so. (c) What fraction of trips go from A to B in the long run? (d) Trips starting in A, B, C earn £10, £8, £12. What is the long-run average fare per trip? (e) For the rotation on $\mathbb{Z}_4$ of C18(e), the long-run fraction of time at each state is $\frac{1}{4}$ but $\mathbb{P}_0(X_n = 0)$ does not converge. Which hypothesis of the convergence theorem fails, and why does the ergodic theorem not need it?

- C22. The fundamental matrix. (a) Fair ruin chain on $\{0, \dots, 5\}$, transient states 1, 2, 3, 4. Write $\mathbf{Q}$ and $\mathbf{R}$, compute $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$, and check $\mathbf{N}\mathbf{1}$ against $i(5 - i)$ and the target column of $\mathbf{NR}$ against $i/5$. Starting from £1, how many times on average is the fortune £3 before the game ends, and how many times £1 (counting the start)? (b) Tennis, $p = 0.6$ (C19). Compute $\mathbf{N}$ for the transient states D, I, O and confirm $h_D = \frac{9}{13}$ and $k_D = \frac{50}{13}$ from $\mathbf{NR}$ and $\mathbf{N}\mathbf{1}$. Interpret $\mathbf{N}_{DD}$ and check it against the visits criterion $\mathbf{N}_{ii} = 1/(1 - f_i)$ with $f_D = \mathbb{P}_D(\text{return to D})$.